

WEEK 5 – PAYMENT TRANSACTION MANAGEMENT SYSTEM

Secenario: The **Payment Transaction Management System** is used to store and manage customer payment transaction details. It maintains payment information related to each order.

PAYMENT_ENTITY: The **PAYMENT_ENTITY** table stores customer payment transaction details.

Attributes:

- **PAYMENT_ID** – Unique payment identifier
- **ORDER_ID** – Order reference
- **PAYMENT_DATE** – Date of transaction
- **PAYMENT_MODE** – UPI / Card / Net Banking / Cash
- **PAYMENT_STATUS** – Success / Failed / Pending
- **TRANSACTION_AMOUNT** – Paid amount

```
SQL> CREATE TABLE PAYMENT_ENTITY (  
2     PAYMENT_ID INT,  
3     ORDER_ID INT,  
4     PAYMENT_DATE DATE,  
5     PAYMENT_MODE VARCHAR2(20),  
6     PAYMENT_STATUS VARCHAR2(20),  
7     TRANSACTION_AMOUNT NUMBER  
8 );  
  
Table created.
```

Establish Relationships and Constraints :

One Order can have **one Payment Transaction**.

The relationship between **ORDER_ENTITY** and **PAYMENT_ENTITY** is **One to One (1 → 1)**.

PRIMARY KEY:

The following Primary Key is added:

- **PAYMENT_ID** – Primary Key of **PAYMENT_ENTITY** table.

A Primary Key uniquely identifies each record in a table.

```
SQL> ALTER TABLE PAYMENT_ENTITY  
2 ADD CONSTRAINT PK_PAYMENT  
3 PRIMARY KEY (PAYMENT_ID);  
  
Table altered.
```

FOREIGN KEY:

The following Foreign Key is used in the **PAYMENT_ENTITY** table:

- **ORDER_ID** references **ORDER_ENTITY(ORDER_ID)**

This Foreign Key establishes the relationship between the two tables.

```
SQL> ALTER TABLE PAYMENT_ENTITY
2 ADD CONSTRAINT FK_PAYMENT_ORDER
3 FOREIGN KEY (ORDER_ID)
4 REFERENCES ORDER_ENTITY(ORDER_ID);
```

Table altered.

NOT NULL Constraint:

NOT NULL ensures that a column cannot contain an empty (NULL) value.

✚ PAYMENT_MODE
✚ PAYMENT_STATUS

```
SQL> ALTER TABLE PAYMENT_ENTITY
2 MODIFY PAYMENT_MODE NOT NULL;
```

Table altered.

```
SQL> ALTER TABLE PAYMENT_ENTITY
2 MODIFY PAYMENT_STATUS NOT NULL;
```

Table altered.

CHECK Constraint:

CHECK validates that the values in a column satisfy a specified condition.

- TRANSACTION_AMOUNT > 0 – Ensures the transaction amount is positive.

```
SQL> ALTER TABLE PAYMENT_ENTITY
2 ADD CONSTRAINT CHK_TRANSACTION_AMOUNT
3 CHECK (TRANSACTION_AMOUNT > 0);
```

Table altered.

INSERT VALUES FOR PAYMENT ENTITY:

```
SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY001', 1001, TO_DATE('20-09-2026','DD-MM-YYYY'), 'UPI', 'SUCCESS', 1500);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY002', 1002, TO_DATE('21-09-2026','DD-MM-YYYY'), 'CARD', 'SUCCESS', 2200);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY003', 1003, TO_DATE('22-09-2026','DD-MM-YYYY'), 'NET BANKING', 'SUCCESS', 1800);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY004', 1004, TO_DATE('23-09-2026','DD-MM-YYYY'), 'UPI', 'SUCCESS', 2500);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY005', 1005, TO_DATE('24-09-2026','DD-MM-YYYY'), 'CARD', 'PENDING', 1200);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY006', 1006, TO_DATE('25-09-2026','DD-MM-YYYY'), 'UPI', 'SUCCESS', 1700);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY007', 1007, TO_DATE('26-09-2026','DD-MM-YYYY'), 'CARD', 'SUCCESS', 1900);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY008', 1008, TO_DATE('27-09-2026','DD-MM-YYYY'), 'NET BANKING', 'SUCCESS', 2300);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY009', 1009, TO_DATE('28-09-2026','DD-MM-YYYY'), 'UPI', 'SUCCESS', 1400);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY010', 1010, TO_DATE('29-09-2026','DD-MM-YYYY'), 'CARD', 'SUCCESS', 2800);
1 row created.
```

```
SQL> SELECT * FROM PAYMENT_ENTITY;
```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
TRANSACTION_AMOUNT				
PAY001	1001	20-SEP-26	UPI	SUCCESS
	1500			
PAY002	1002	21-SEP-26	CARD	SUCCESS
	2200			
PAY003	1003	22-SEP-26	NET BANKING	SUCCESS
	1800			
PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
TRANSACTION_AMOUNT				
PAY004	1004	23-SEP-26	UPI	SUCCESS
	2500			
PAY005	1005	24-SEP-26	CARD	PENDING
	1200			
PAY006	1006	25-SEP-26	UPI	SUCCESS
	1700			
PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
TRANSACTION_AMOUNT				
PAY007	1007	26-SEP-26	CARD	SUCCESS
	1900			
PAY008	1008	27-SEP-26	NET BANKING	SUCCESS
	2300			
PAY009	1009	28-SEP-26	UPI	SUCCESS
	1400			
PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
TRANSACTION_AMOUNT				
PAY010	1010	29-SEP-26	CARD	SUCCESS
	2800			

10 rows selected.

REPORT 1 :

Display all successful payments

```
SQL> SELECT *  
2 FROM PAYMENT_ENTITY  
3 WHERE PAYMENT_STATUS = 'SUCCESS';
```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
TRANSACTION_AMOUNT				
PAY001	1001	20-SEP-26	UPI	SUCCESS
	1500			
PAY002	1002	21-SEP-26	CARD	SUCCESS
	2200			
PAY004	1004	23-SEP-26	UPI	SUCCESS
	2500			
PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
TRANSACTION_AMOUNT				
PAY006	1006	25-SEP-26	UPI	SUCCESS
	1700			
PAY007	1007	26-SEP-26	CARD	SUCCESS
	1900			
PAY008	1008	27-SEP-26	NET BANKING	SUCCESS
	2300			
PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
TRANSACTION_AMOUNT				
PAY009	1009	28-SEP-26	UPI	SUCCESS
	1400			
PAY010	1010	29-SEP-26	CARD	SUCCESS
	2800			

8 rows selected.

Failed Transactions

```
SQL> SELECT *
  2 FROM PAYMENT_ENTITY
  3 WHERE PAYMENT_STATUS = 'FAILED';
```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
TRANSACTION_AMOUNT				
PAY003	1003	22-SEP-26	NET BANKING	FAILED
	1800			

```
SQL> SELECT *
```

Count total successful and failed payments

```
SQL> SELECT PAYMENT_STATUS,
  2 COUNT(*) AS TOTAL_PAYMENTS
  3 FROM PAYMENT_ENTITY
  4 WHERE PAYMENT_STATUS IN ('SUCCESS', 'FAILED')
  5 GROUP BY PAYMENT_STATUS;
```

PAYMENT_STATUS	TOTAL_PAYMENTS
FAILED	1
SUCCESS	8

Update failed payments after retry

```
SQL> UPDATE PAYMENT_ENTITY
  2 SET PAYMENT_STATUS = 'SUCCESS'
  3 WHERE PAYMENT_ID = 'PAY003'
  4 AND PAYMENT_STATUS = 'FAILED';

1 row updated.
```

Identify pending transactions

```
SQL> SELECT *
  2 FROM PAYMENT_ENTITY
  3 WHERE PAYMENT_STATUS = 'PENDING';
```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
TRANSACTION_AMOUNT				
PAY005	1005	24-SEP-26	CARD	PENDING
	1200			

REPORT :2

Number of transactions using UPI

```
SQL> SELECT COUNT(*) AS UPI_TRANSACTIONS
  2 FROM PAYMENT_ENTITY
  3 WHERE PAYMENT_MODE = 'UPI';
```

UPI_TRANSACTIONS
4

Number of card payments

```
SQL> SELECT COUNT(*) AS CARD_PAYMENTS
  2 FROM PAYMENT_ENTITY
  3 WHERE PAYMENT_MODE = 'CARD';
```

CARD_PAYMENTS
4

Most preferred payment method

```
SQL> SELECT PAYMENT_MODE,
2          COUNT(*) AS TOTAL_TRANSACTIONS
3 FROM PAYMENT_ENTITY
4 GROUP BY PAYMENT_MODE
5 ORDER BY TOTAL_TRANSACTIONS DESC;
```

PAYMENT_MODE	TOTAL_TRANSACTIONS
CARD	4
UPI	4
NET BANKING	2

REPORT :3

Total revenue generated

```
SQL> SELECT SUM(TRANSACTION_AMOUNT) AS TOTAL_REVENUE
2 FROM PAYMENT_ENTITY
3 WHERE PAYMENT_STATUS = 'SUCCESS';
```

TOTAL_REVENUE
18100

Revenue by payment method

```
SQL> SELECT PAYMENT_MODE,
2          SUM(TRANSACTION_AMOUNT) AS REVENUE
3 FROM PAYMENT_ENTITY
4 WHERE PAYMENT_STATUS = 'SUCCESS'
5 GROUP BY PAYMENT_MODE;
```

PAYMENT_MODE	REVENUE
NET BANKING	4100
CARD	6900
UPI	7100

Average transaction amount

```
SQL> SELECT AVG(TRANSACTION_AMOUNT) AS AVERAGE_TRANSACTION_AMOUNT
2 FROM PAYMENT_ENTITY
3 WHERE PAYMENT_STATUS = 'SUCCESS';
```

AVERAGE_TRANSACTION_AMOUNT
2011.11111

REPORT :4

```

SQL> SELECT C.CUSTOMER_NAME,
2         O.ORDER_ID,
3         O.ORDER_DATE,
4         P.PAYMENT_MODE,
5         P.TRANSACTION_AMOUNT AS AMOUNT,
6         P.PAYMENT_STATUS
7 FROM CUSTOMER C
8 JOIN ORDER_ENTITY O
9 ON C.CUSTOMER_ID = O.CUSTOMER_ID
10 JOIN PAYMENT_ENTITY P
11 ON O.ORDER_ID = P.ORDER_ID
12 ORDER BY P.PAYMENT_ID;

```

CUSTOMER_N	ORDER_ID	ORDER_DAT	PAYMENT_MODE	AMOUNT
PAYMENT_STATUS				

ARUN SUCCESS	1001	20-SEP-26	UPI	1500
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PRIYA SUCCESS	1002	21-SEP-26	CARD	2200
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KARTHIK SUCCESS	1003	22-SEP-26	NET BANKING	1800
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CUSTOMER_N	ORDER_ID	ORDER_DAT	PAYMENT_MODE	AMOUNT
PAYMENT_STATUS				

DIVYA SUCCESS	1004	23-SEP-26	UPI	2500
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SURESH PENDING	1005	24-SEP-26	CARD	1200
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MEENA SUCCESS	1006	25-SEP-26	UPI	1700
------------------	------	-----------	-----	------

CUSTOMER_N	ORDER_ID	ORDER_DAT	PAYMENT_MODE	AMOUNT
PAYMENT_STATUS				

VIGNESH SUCCESS	1007	26-SEP-26	CARD	1900
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AMU SUCCESS	1008	27-SEP-26	NET BANKING	2300
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RAHUL SUCCESS	1009	28-SEP-26	UPI	1400
------------------	------	-----------	-----	------

CUSTOMER_N	ORDER_ID	ORDER_DAT	PAYMENT_MODE	AMOUNT
PAYMENT_STATUS				

NISHA SUCCESS	1010	29-SEP-26	CARD	2800
------------------	------	-----------	------	------

10 rows selected.